

Experiment 5: Non-Preemptive Priority

C Program

```
#include<stdio.h>

int main()
{
    int bt[20],pr[20],p[20],wt[20],tat[20];
    int i,j,n,temp;

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        p[i]=i+1;
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
        printf("Enter Priority for P%d: ",i+1);
        scanf("%d",&pr[i]);
    }

    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(pr[i]>pr[j])
            {
                temp=pr[i]; pr[i]=pr[j]; pr[j]=temp;
                temp=bt[i]; bt[i]=bt[j]; bt[j]=temp;
                temp=p[i]; p[i]=p[j]; p[j]=temp;
            }

    wt[0]=0;
    for(i=1;i<n;i++) wt[i]=wt[i-1]+bt[i-1];
    for(i=0;i<n;i++) tat[i]=wt[i]+bt[i];

    printf("\n\nProcess\\tPriority\\tBT\\tWT\\tTAT\\n");
    for(i=0;i<n;i++)
        printf("P%d\\t%d\\t%d\\t%d\\t%d\\n",p[i],pr[i],bt[i],wt[i],tat[i]);

    return 0;
}
```

Output

Sample Input:

```
4
5 2
3 1
8 4
6 3
```

Sample Output:

```
Process Priority BT WT TAT
P2 1 3 0 3
P1 2 5 3 8
P4 3 6 8 14
P3 4 8 14 22
```